

Lilly George

412-865-8732 | georg211@purdue.edu | www.linkedin.com/in/lilly-george | <https://lillygeorge.pages.dev>

EDUCATION:

Purdue University John Martinson Honors College: *West Lafayette, IN*

Class of 2028

Bachelor of Science, Mechanical Engineering, Minor Materials Science and Engineering

GPA 3.4

EXPERIENCE:

Honda Motor Company – Mechanical Engineering Test Intern, *Haw River, NC*

May 2026 – August 2026

- Collected endurance test strain/acceleration data from lawnmower, calculated damage accrued in MATLAB, and correlated data to hydraulic actuator movement, condensing test time by half and increasing efficiency
- Modeled (CATIA) and CAE-validated a lawnmower blade plate redesign that eliminated yielding during bolt torquing, further validated through physical stress analysis; presented solution and resolved yielding issue
- Constructed drive system and circuit board for troubleshooting electric motor hall sensors and thermistors
- Performed thermal study on prototype; ensured engine matching criteria met, compared to competitors

Purdue Formula SAE Electric Racing – Vehicle Dynamics, *Purdue University*

August 2024 – Present

- Designed and set up Ansys FEA for 16 suspension tabs; cut on rotary tube laser and created jig for welding
- Analyzed carbon fiber suspension members in uniaxial tension to refine bonding parameters for strength
- Created MATLAB script to optimize anti-roll bar geometry/material/heat treat based on stiffness targets

Undergraduate Research Assistant, *Purdue Layered Materials and Structures Lab*

January 2026 – Present

- Investigated feasibility/properties of rare earth metal doping in MXenes for semiconductive applications

Cummins – Mechanical Engineering Intern, *Columbus, IN*

June 2025 – August 2025

- Followed and supported a fast-paced 60L genset program; determined necessary engine validation tasks
- Researched turbomachinery integration on larger alternative fuel (biodiesel, H₂) engines
- Broke down turbocharger load limit calculations to improve prior practices and push product performance

LEADERSHIP:

Mentor – Women in Engineering 1:1 Mentoring Program, *Purdue University*

August 2025 – Present

- Monthly meetings with freshman mentee to share advice and empower to ensure success in college

Volunteer - Pittsburgh Vintage Grand Prix, *Pittsburgh, PA*

June 2015 – June 2024

- Volunteered for three 16-hr days each summer at car shows; oversaw registration/parking of 300+ cars/day
- Foundation raises \$300,000/year for charities benefiting those with Autism

STUNT Team Member, *Purdue University*

August 2024 – May 2025

- Helped formulate inaugural team, attended two tournaments and contributed to first game victory
- Practiced nine hours a week to learn 15 stunting and tumbling routines from August to February

Moon Area Gymnastics Association – Coach, *Moon Township, PA*

December 2019 - May 2024

- Led 15-20 young gymnasts as part of a team to teach skill techniques and safety through patience, positive reinforcement, verbal corrections, and physical demonstrations to improve their gymnastics abilities
- Choreographed/taught a 20 diverse gymnasts a 3-minute routine to highlight their skills in a showcase

SKILLS AND ACHIEVEMENTS:

Skills: Siemens NX/Teamcenter, MATLAB, UTM Testing, Manual Mill, Manual Lathe, Waterjet, Rotary Tube Laser, Strain Gauges, Accelerometers, Thermocouples, Ansys FEA, Autodesk Inventor, Fusion360, Catia V5, Python

Achievements: Dean's List/Semester Honors (x2/x3), Alpha Lambda Delta/Phi Eta Sigma Honors Society member, John Martinson Honors College student, Commended by National Merit Scholarship Program (2024)